Glossary of terms

The words of every session, with the plain meaning. Learn the industry term; the plain words explain it.

Term	Plain meaning	Session
Adam	An optimiser (Kingma and Ba, 2014) that gives each parameter its own step size.	10
Attention	Each token takes a weighted average of the value vectors, with weights from softmax(Q K^T / sqrt(d)).	18
Backpropagation	Computing the derivatives of the loss by applying the chain rule backwards through the computation graph.	9
Backpropagation	The backward pass that applies the chain rule from the loss back to every parameter, reusing the values stored in the forward pass.	11
Backpropagation	The chain rule applied backwards through a network to get the gradient of the loss for every weight.	18
Backward pass	Computing the gradient of the loss for every parameter, from the output back to the input.	11
Bayes' theorem	The rule P(H | E) = P(E | H) P(H) / P(E): the posterior from the likelihood, the prior and the evidence.	15
Bias and variance	Bias: the error of the average prediction. Variance: how much the prediction moves between training sets.	16
Binomial distribution	The number of successes in n independent yes or no trials, each with probability p.	14
Central limit theorem	Averages of many independent values are close to Gaussian, whatever the original shape.	14
Chain rule	The derivative of a composite function is the derivative of the outer function times the derivative of the inner function.	9
Composite function	A function of a function: the output of one function becomes the input of the next. Written f(g(x)).	9
Computation graph	A picture of a calculation as boxes (values) and arrows (operations), from the inputs to the loss.	9
Conditional independence	Two words are independent within each class (spam, ham), though not overall. Independence itself is from Session 13.	15
Conditional probability	P(A | B): the probability of A once we know B happened.	15
Contour plot	A map of a function of two inputs. Each curve joins the points where the function has the same value.	8
Cosine similarity	The cosine of the angle between two vectors: 1 for the same direction, 0 at a right angle.	2
Covariance and correlation	How two columns move together, and the same thing scaled to lie between -1 and +1.	13
Covariance matrix	A table of how every pair of features varies together. The diagonal holds each feature's variance, the typical squared distance from its mean.	5
Cross-entropy	The average surprise when reality follows P but the model believes Q. The training loss for classification and language models.	17
Delta	The gradient of the loss with respect to the input of a unit, written with the Greek letter delta.	11

Term	Plain meaning	Session
Derivative	The slope of a function at one point: how fast the output changes when the input changes. Written f-prime of x or df/dx.	7
Determinant	One number from a square matrix. Its size is the factor by which A scales areas.	4
Dot product (inner product)	Multiply matching entries of two vectors, then add. The result is one number.	2
Eigenvalue	The factor by which an eigenvector is stretched. Written with the Greek letter lambda (λ).	5
Eigenvector	A direction that a matrix only stretches, flips or squashes. It stays on its own line.	5
Embedding	A vector that represents a piece of text so that similar meanings get similar vectors.	2
Entropy	The average surprise of a distribution, in bits with log2 or nats with ln. How uncertain the outcome is.	17
Epoch and mini-batch	An epoch is one pass through all the training rows. A mini-batch is a small group of rows used for one step.	10
Expectation, variance and standard deviation	The probability-weighted average mu, the average squared distance from it, and its square root sigma.	13
Exponential function	A function where the input is a power, such as 2^x (2 to the power x). It grows by repeated multiplying.	1
Fine-tuning and LoRA	Fine-tuning: training an already trained model a little more on new data (training is taught in Sessions 7 to 10). LoRA (low-rank adaptation) does it with two thin matrices.	6
Finite difference	An estimate of a slope from two nearby points: change in output divided by a small change in input.	7
Forward pass	Computing the output and the loss from the inputs, layer by layer.	11
Function	A rule that turns each input number into one output number, written f(x).	1
Gaussian (normal) distribution	The bell curve, set by its mean mu and standard deviation sigma.	14
Gradient	The vector of all the partial derivatives, written as a column. It points where the function increases fastest. Symbol: nabla.	8
Gradient descent	Repeatedly changing the parameters a little in the direction that lowers the loss.	10
Identity matrix	A square matrix of 1s on the diagonal and 0s elsewhere. It leaves every vector unchanged.	3
Independent random variables	Knowing the value of one does not change the probabilities of the other, like two separate dice.	13
Inverse	The matrix A^{-1} that undoes A: A^{-1} times A is the identity matrix.	4
Jacobian	The matrix of all the partial derivatives of a function with several inputs and several outputs. One row for each output, one column for each input.	9
KL divergence	Kullback-Leibler divergence: cross-entropy minus entropy. The extra cost of believing Q instead of P.	17
L2 norm (Euclidean norm)	The length of a vector: the square root of the sum of its squared entries.	2
L2 penalty (ridge) and weight decay	Lambda times the sum of the squared weights, added to the loss. In plain gradient descent it shrinks every weight a little at each update.	12

Term	Plain meaning	Session
Learning rate (eta)	The size of each step. We write it as the Greek letter eta; some books write alpha.	10
Least squares	Choosing the line whose squared errors add up to the smallest total.	4
Likelihood	How probable the exact data we saw are, for one chosen value theta of the parameter. Written p(data | theta).	16
Linear system	Several equations in the same unknowns, written together as Ax = b.	4
Log-likelihood	l(theta) = ln p(data | theta), in nats. A product becomes a sum. (L stays the loss.)	16
Log-probability	The natural logarithm of a probability. It is zero or negative, because a probability is at most 1.	1
Log-sum-exp	m + ln(sum of e^(z - m)) with m the largest score: the log of a sum of exponentials, computed safely.	12
Logarithm (ln)	The inverse of an exponential. The natural logarithm ln uses the base e ≈ 2.71828.	1
Logistic regression	A model that passes a straight-line score through the sigmoid to give a probability between 0 and 1.	18
Loss function	One number that says how wrong a model is on the data. Smaller is better. Here: the mean squared error.	7
Low-rank approximation	A matrix rebuilt from only its first few singular values. It needs far fewer numbers to store.	6
Matrix	A table of numbers with m rows and n columns. Its shape is m × n.	3
Matrix multiplication	A rule that combines two matrices. Each entry of the answer is a dot product of a row and a column.	3
Maximum likelihood estimation (MLE)	Choose the parameter value with the largest likelihood.	16
Minimum	A lowest point of a curve. At a minimum, or a maximum, the slope is zero.	7
Momentum	A running memory of past gradients that smooths the path and speeds it up.	10
Naive Bayes classifier	A classifier that multiplies per-word likelihoods, assuming the words are independent within each class.	15
Overflow and underflow	A result too large or too small for the number format: it becomes inf or 0.	12
Parameter	A number inside a model that training can change. Our line y-hat = b + w x has two: the intercept b and the slope w.	8
Parameters	The numbers inside a model that training sets: its weights W and biases b.	3
Partial derivative	The slope of a function when you change one input and hold the others fixed. Written ∂L/∂w and read "partial L, partial w".	8
Perplexity	e to the power of the mean negative log-probability. The effective number of equally likely choices the model has at each word.	17
Power iteration	Multiplying by a matrix again and again to find its top eigenvector.	5
Principal component	A direction of largest spread in the data: an eigenvector of the covariance matrix.	18
Principal component analysis (PCA)	Finding the directions of greatest spread in data, then keeping only the first few.	6
Prior and MAP	The prior is what we believe before the data. Maximum a posteriori (MAP) estimation maximises likelihood times prior.	16

Term	Plain meaning	Session
Prior, likelihood and posterior	The belief before evidence, the chance of the evidence if the hypothesis is true (for a test: its sensitivity), and the updated belief.	15
Probability mass function (pmf) and density function (pdf)	The pmf gives P(X = x) for values you can list. The pdf is a curve, and probability is the area under it.	13
Query, key, value	The three vectors each token makes in attention: what it looks for, what it offers and what it passes on.	18
Random variable	A rule that turns the outcome of a chance event into a number. We write it X.	13
Rank	How many independent directions the columns of a matrix give: no chosen column can be built from the others.	4
Regularisation	Any change to training that discourages a model from memorising its training data.	12
Sampling	Drawing a value at random according to its probabilities, as a chat model draws its next word.	14
Sigma notation	The symbol Σ means add up. Sigma notation is a short way to write a long sum.	1
Sigmoid and tanh	Sigmoid 1 / (1 + e^-z) squeezes any number into 0 to 1; tanh squeezes it into -1 to 1.	11
Singular value	A stretch factor in Σ. The singular values are listed from largest to smallest.	6
Singular value decomposition (SVD)	Any matrix written as rotate, stretch, rotate: $A = U\Sigma V^T$.	6
Softmax	Turns a list of scores (logits) into probabilities that are positive and add up to 1.	12
Standard error, confidence interval, p-value	How much a sample mean wanders, a range built from it, and how surprising a gap would be if nothing changed.	14
Steepest increase	The direction in which a function rises fastest. Its opposite, minus the gradient, is the direction of steepest decrease.	8
Tangent line	The straight line that just touches a curve at one point. Its slope is the derivative there.	7
Temperature and top-p	Sampling settings. Temperature divides the logits before the softmax. Top-p keeps only the most likely words.	17
Trace	The sum of the diagonal entries of a square matrix.	5
Transpose	The matrix flipped so that rows become columns. It is written with a small T.	3
Vector	A list of d numbers in a fixed order. A row of data is a vector. d is its dimension.	2